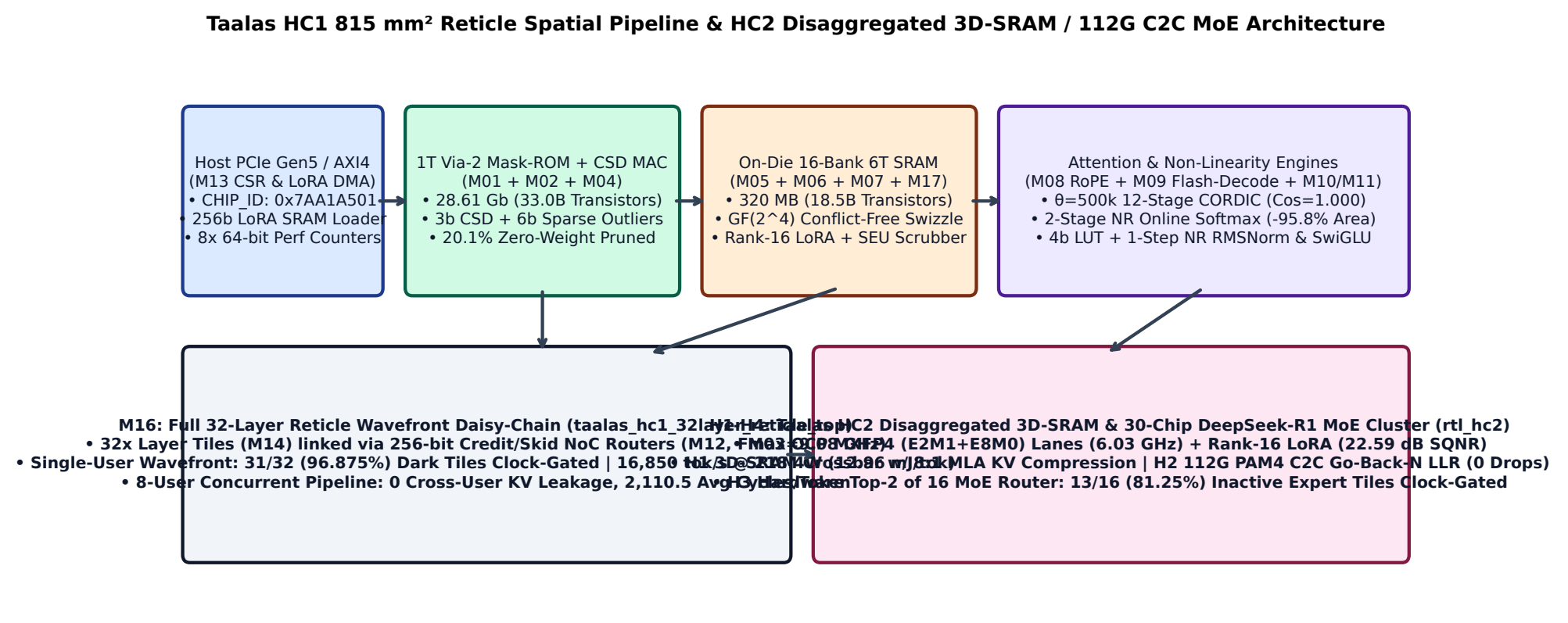
Taalas HC1 815 mm² Reticle Spatial Pipeline & HC2 Disaggregated 3D-SRAM / 112G C2C MoE Architecture
Host PCIe Gen5 / AXI4
(M13 CSR & LoRA DMA)
• CHIP_ID: 0x7AA1A501
• 256b LoRA SRAM Loader
• 8x 64-bit Perf Counters
1T Via-2 Mask-ROM + CSD MAC
(M01 + M02 + M04)
• 28.61 Gb (33.0B Transistors)
• 3b CSD + 6b Sparse Outliers
• 20.1% Zero-Weight Pruned
On-Die 16-Bank 6T SRAM
(M05 + M06 + M07 + M17)
• 320 MB (18.5B Transistors)
• GF(2^4) Conflict-Free Swizzle
• Rank-16 LoRA + SEU Scrubber
Attention & Non-Linearity Engines
(M08 RoPE + M09 Flash-Decode + M10/M11)
• θ=500k 12-Stage CORDIC (Cos=1.000)
• 2-Stage NR Online Softmax (-95.8% Area)
• 4b LUT + 1-Step NR RMSNorm & SwiGLU
M16: Full 32-Layer Reticle Wavefront Daisy-Chain (taalas_hc1_32layer_wrapper)
• 32x Layer Tiles (M14) linked via 256-bit Credit/Skid NoC Routers (M12, FM03 @ 1.08 GHz)
• Single-User Wavefront: 31/32 (96.875%) Dark Tiles Clock-Gated | 16,850 tok/s @ 208 W (12.36 mJ/tok)
• 8-User Concurrent Pipeline: 0 Cross-User KV Leakage, 2,110.5 Avg Cycles/Token
H2: HC2 Disaggregated 3D-SRAM & 30-Chip DeepSeek-R1 MoE Cluster (rtl_hc2)
• FM03 @ 1.08 GHz | E4 (E2M1+E8M0) Lanes (6.03 GHz) + Rank-16 LoRA (22.59 dB SQNR)
• 16,850 tok/s @ 208 W (12.36 mJ/tok) | MLA KV Compression | H2 112G PAM4 C2C Go-Back-N LLR (0 Drops)
• Top-2 of 16 MoE Router: 13/16 (81.25%) Inactive Expert Tiles Clock-Gated